

ActInf GuestStream #023.1 ~ Maxwell Ramstead, Dalton Sakthivadivel, et al

GuestStream | Maxwell Ramstead, Dalton Sakthivadivel, et al. | 1:39:38

Hello and welcome everyone. It is June 13th, 2022, and we're here in Acton Flab guest stream number 23.1 with Maxwell and Dalton. We're talking about rebooting the free energy principle literature and their recent publication. So I'll pass it to the two of you to provide just an initial introduction and some context for the paper before we jump into some specific parts.

Maxwell first. Cool. My name is Maxwell Ramstad. I'm the director of research at the Versus Research Lab, where we specialize in probabilistic approaches to artificial intelligence. And of late, most of my research has been on the physics foundations of the free energy principle. And yeah, very, very much looking forward to having this conversation.

Awesome. And Dalton?

Sure. Okay. Well, I'm Dalton. I'm a mathematician and a physicist. So right now I'm at Stony Brook University in New York. I'm also at the Versus Lab.

Where we're doing some work on the free energy principle. Right now my research interests are about formal approaches to biological physics and mathematical foundations of complex systems and these kinds of questions. One of the most interesting frameworks in this sphere is the free energy principle. And that's how I've gotten involved in this

literature. Awesome. So maybe in the same order, what brought us to this paper? What was the gap that this paper stepped into? And how did this team arise around the paper? And I'll bring it up on the screen as well.

Good question. Well, I think the paper does a bunch of different things. The first, I mean, we've presented this as rebooting the free energy principle literature. I think, yeah, over the last year, we've presented this as rebooting the free energy principle literature.

I think, yeah, over the last, it's been almost two decades now since the FUP has been a thing. I think the first paper was circa 2005 ish. And the framework, or the field of study that is emerged around the free energy principle has changed a lot over the last two decades, no small part in terms of the self understanding of the free energy principle.

And the understanding that we have as a community of practitioners working with these techniques. And I mean, as Dalton pointed out a while back, I think the last time that really, the FUP literature had been surveyed in a systematic fashion to kind of present where we were at, where we are currently like the state of the art, I was probably called

the FUP monograph from the FUP monograph from 2019 January 2019. I mean, you know, both of you were part of this international physics reading group that we had, this was not, this was not a simple piece to read. And yeah, since then, there hasn't really been basically a summary of where the literature is. So on the one hand, we were aiming to reboot the literature.

And to basically summarize the ways that the free energy principle had been applied in the literature and to also present a lot of the progress that had been made over the last three years, a lot of which is due to Dalton.

So I think that segues nicely into the Dalton.

Well, too generous by far as usual. But yeah, so Maxwell raises a good point that there hasn't been really an effort to tie together all the different threads since probably 2019.

And one of the reasons why, I mean, that's, that's a long time, but it's not overly long. However, in the last couple of years, there have been some pretty serious advances in the state of the mathematics and the physics behind the free energy principle.

So now, you know, three years later, we're in 2022. It's not a long time, but nevertheless, there have been big developments that really beg this kind of a survey to be done.

And so that was one of the aims, at least of the paper was to draw together all the different threads, to unify them, make them a bit more cohesive, and present the free energy principle as we're conceptualizing it today, in a very top to bottom fashion.

So before we jump into the sections and the arguments and the claims in the paper, which I'm sure will cover a lot of these updates, I'd just love to hear what is the free energy principle? What is active inference? And how are they related?

I mean, I guess, in the most general sense, the free energy principle is a statement that says that we can understand the dynamics of self-organizing systems, as following a path of least surprisal, in the most general sense.

And then in that context, it's good that you asked the question, I think that there are at least two uses of the term active inference in the literature.

The one that I've tended to use is a very broad sense, which is just like active inference is specifying a path of least action for what are called autonomous states.

So like the active and internal states of systems that are considered things under the free energy formulation. Active inference has also been used to refer to a class of specific partially observable Markov decision process models that have been deployed.

And, you know, most of the in silico implementations of the free energy principle rely on these.

I think this has led to a little bit of confusion in that, you know, when folks say active inference is a general theory, able to account for self-organization, they're not saying this specific class of POM-VP models is effectively repurposable for everything.

So, yeah, I would say that this is the first sense that I outlined is the sense in which we say that active inference is a corollary of the free energy principle.

And yeah, I mean, in its most general form, the free energy principle is just a statement, an alternative statement or a restatement of the principle of stationary action where the action is defined as a surprisal in effect.

So, yeah.

Dalton, same questions to you.

Yeah, I think Maxwell pretty much covered it.

So as I mentioned, you know, my background is actually not in the free energy principle, not originally, and it's not even in machine learning.

So I have come at this literature with a bit of a different set of ideas and traditions and tools.

And so as a result, I can't really talk about active inference.

I'm not totally familiar with it, but I can talk about the free energy principle at length.

I'll try to be brief.

So to me, the free energy principle is a statement that random dynamical systems do, on average, unsurprising things.

And it's a very simple statement.

All it means is that if you have a random dynamical system and it's trying to get to some place in state space, that is not surprising to it.

So some preferred region of state space, maybe a particular phenotype, maybe a particular morphological blueprint, something like this.

The free energy principle is a statement of not only that that happens, but also a set of tools to model how that happens.

How do systems get to these kinds of attractors?

And there's this very interesting way in which the getting to that attractor can be read as inference.

And this is where all these connections in literature appear to things like active inference.

You can talk about how systems actually get themselves to that attractor as a perception and action problem.

And you can model it using this very famous variational free energy bound.

And you can show, okay, when the system is encoding some sort of belief about its environment and about itself, that when it does this inference, you have the minimization of some upper bound unsurprisal.

So the system is at least doing unsurprising things at the level of, you know, it's not exceeding some kind of intrinsic surprisal.

That's some of the early formulations.

Some of the more recent formulations take it even further.

And you say the surprisal, full stop of some trajectory of the system is going to be minimized by the system.

So it's going to take, on average, the system is going to do what it expects to do, which in this case is get to some region in the state space that it prefers to inhabit.

What makes that not a tautology and what makes it something that we can actually operationalize and ask interesting questions about is the fact that there is this nice connection to inference.

There are all of these explicit mathematical tools that we can use to talk about surprisal minimization, not just as the kind of information theoretic statement of systems that aren't surprising or systems that aren't surprised do unsurprising things.

It ends up a little bit more nuanced than that.

And so to me, the free energy principle is this whole host of results about random dynamical systems and especially random dynamical systems which are organized or which self-organize.

I think that is how I would think of it.

And it's got some physics and stuff that we can actually talk about in that context.

Awesome. Thanks both for those great answers.

I'm pulling up the paper and so many things in what you said I just wanted to follow up on, but let's follow the main line of the paper and wherever you think there's some dots that could be connected or any questions arising.

Or areas of future research or the connection to applications that you're working on or other people might think about getting involved in.

We can just take those little cul-de-zacs from the main line.

So I have the paper up on Bayesian mechanics, a physics of and by beliefs.

The two of you are the first authors.

There's also Connor Hines, Magnus Kudall, Baron Milledge, Lancelot DaCosta, Brennan Klein and Carl Friston.

Any just overview notes on again like what led to this team or anything like that while we're on this first page?

Well, from my perspective, Dalton and I have been talking a lot about this kind of stuff for the last year, year and a half, two years.

And yeah, I won't, I won't try to, I'll try not to be too overly generous, Dalton, but like, I think, yeah.

So some of Dalton's work has, I think, changed the way that we think about the FEP.

In particular, what Dalton did in this great paper towards a geometry and analysis for Bayesian mechanics is to show that the free energy principle and the principle of constrained

and axiom entropy are actually dual to each other, they are two sides of the same coin.

And I, you know, I've been following this line of work for quite some time, and I think it foundationally changes the way that we think about the FEP.

It moves it from like this, some might say exotic result from, you know, theoretical neurobiology to, you know, it connects it back up to the foundations of contemporary physics effectively by showing that, you know, this thing that we understand and almost take for granted, you know, maximum entropy inference procedures turn out to connect up to this free energy literature in a non-trivial way.

So I think the paper started out, and it didn't end this way, but I think we started off trying to write a reader friendly companion to the work that Dalton had done.

But by the end of it, realized that this was more, I think that Dalton's paper towards geometry and analysis for Bayesian mechanics,

pillars, I think it stands on its own as a piece of mathematical reasoning.

So it is an exercise in maximum entropy inference math, and it shows that you can recover some of the really core results of the free energy formulation, just using the language of constraint, maximum entropy.

In particular, Dalton approves the approximate Bayesian inference lemma that had caused so much discussion in our literature for the last bit.

And so, you know, I think the paper grew out of that.

And yeah, the team is something of a who's who of like, you know, the core contributors to this literature.

And just from a strategic point of view, it was important for us to have like, you know, Lance DaCosta, Karl Friston, you know, Connor Hines and folks that had been, I think, contributing to this core formulation and in particular to the development of, you know, this new framing of the free energy stuff, Bayesian mechanics.

So yeah, I think that the paper grew out of that.

And if Dalton's paper is this Dalton's stint, like, towards the geometry and analysis for Bayesian mechanics is a kind of standalone piece of mathematical reasoning.

This is more about how this actually fits into the free energy principle literature.

So this is very much more a free energy principle literature paper.

And yeah, we were hoping to do, you know, as reader friendly and introduction as we could.

There are actually two whole kind of introductory sections in the paper.

But we ended up doing a lot of, I think, important work in that paper connecting up to the free energy principle.

I hope that helps answer me.

Great answer and welcome, Lars.

Would you like to give any overviews or notes before we jump into the sections of the paper?

Hey, thanks.

Thanks for sending me the link there, Max.

I'll let me down here.

No, nothing really to add there.

I'm just here as a spectator.

I have a few questions as we go down along and maybe I can turn into the conversation.

But yeah, thanks for having me along.

Awesome.

All right.

So the abstract begins with a really clear aim.

The aim of this paper is to introduce a field of study that has emerged over the last decade called Bayesian mechanics.

And we're going to be unpacking that claim.

So suffice to say, the abstract is very clear and summarizes some of the core points that we're going to delve into now.

We're looking now at the contents.

So what led to the sections of the paper being this way and in this order?

What did you feel like it was important to include?

What did you feel like it was important to introduce and why?

Well, so if you look, I'll actually pull it up in front of me so that we were all talking about the same thing.

Give me one second.

And it's viewable in large size on the live stream.

So yeah, feel free to have your own copy up, but I'll have whatever section we're looking at will be very visible on the live stream.

While Maxwell is grabbing his copy, I'll just give a couple of contextual remarks.

So like Maxwell described, I mean, originally I had released this set of results about what we call approximate Bayesian inference and the sort of mathematical basis for some of the key ideas in the FEP literature up until that point anyway.

And originally it was meant to be a much shorter paper that just dissected some of those results and related them back to the FEP in a way that was a little more concrete and a way that people who actually work on the FEP might care a little more about.

Because of course, the stuff that I wrote about is a little far removed from the FEP even though it's an object of inspiration for the results in the in that paper.

So there is a reading of that paper where it's formally pretty distinct from the FEP.

So originally that was the goal is just give it some extra physics intuition and some extra connection to the actual FEP.

It became something a little bit different as we were building this out and we realized, okay, there are a whole set of new results, not only mine, but also from Carl's TNB group and from others folks.

And they all really should be tied together nicely.

And part of that is the, you know, connecting the geometry and analysis paper to the actual FEP.

Part of it is actually talking about what we would think of the FEP as today.

That shaped the table of contents a little bit, because a lot of what we then wanted to go and say is, okay, we have this theory of the free energy principle.

One of the key ideas in the way that we're viewing the free energy principle today is that it is just a bit of physics.

It is much like classical mechanics or statistical mechanics.

It has a set of mathematical results that sort of hang around it, that give it a nice foundation, and it applies to specific systems and describes them in a specific way.

And that was why we began with both an overview of what is mechanics, what do we talk about in physics when we say mechanics, and also section three is what is the free energy principle?

What have we talked about so far in the literature, and what might we define as Bayesian mechanics?

So what is the physics of beliefs or systems that carry beliefs?

So that sort of shaped the first maybe 20 pages of the papers.

It's just building out the theory and the context for the theory.

Afterwards is a bit of a sort of change in our line of attack, but you'll see it does remain very similar.

So I won't get too far ahead of myself.

Great.

Any other notes, Maxwell, or we can carry on?

No, I think that's a great summary.

I think we try to keep this as self-contained as possible as a paper.

So, you know, the two, you know, introductory sections are meant to provide some kind of visual intuition for a lot of the more formal results that are discussed in the paper and in the papers that it builds on.

We have, you know, the artistic brilliance of Brennan Klein that's added to this paper.

So, you know, I think the illustrations help.

The idea was that, you know, based on Dalton's work, we are formulating basically a geometry and analysis for the core results in the free energy principle literature.

So we thought, let's leverage the geometric nature of a lot of these intuitions to really illustrate them.

And so, yeah, that I think explains a lot of the structure.

And the paper ends on some philosophical considerations and hopefully clarifies some of the confusions or misunderstandings, understandable misunderstandings that have arisen.

So, yeah, so I think that does really address.

I guess the last thing to point out is we introduced this field of study in the last section that we we call G theory, which is, I think, the direction that we're going to be moving in for a while, which is formulating the free

energy principle in the Bayesian mechanics in terms of maximum caliber.

So extending this duality between the density dynamics formulation, the free energy principle in terms of probability densities over states and maximum constraint, maximum entropy to path based formulation.

So path entropy is called caliber and you can doubt them as one of the world specialists in maximum caliber.

And yeah, maximum caliber is maximum path entropy.

So it's the same idea as maximum entropy, but you're considering the entropy of paths.

And we have had, you know, the free energy principle started off as a formulated in terms of paths.

After all, it is a version of the principle of stationary action that leads us to write down paths of stationary action.

stationary action. And so the the ambition now is to connect these things up much more formally so that we can we can handle really genuine non equilibria.

things like wandering Markov blankets and moving attractors and all that kind of stuff, which find the, you know, a home much more naturally in the language of maximum caliber.

Did you did you want to add anything to that?

Dalton, you're really the Max Cal expert.

I think that's a good account of one of the key future directions is, you know, we spend the first half introducing Bayesian mechanics.

We spend the second half talking about where Bayesian mechanics fits into the rest of statistical mechanics and high energy physics by connecting it to Maxent and to gauge theory.

And this is where some of the geometric stuff comes in.

And then the very last thing we do, we sort of cap it off by saying, okay, now, you know, we've talked a lot about Bayesian mechanics.

We've talked a lot about maximum entropy Bayesian mechanics, but there is in some sense, much more natural formulation of Bayesian mechanics in terms of dynamics,

because ultimately, that's what we're thinking about is dynamical systems and the way that they evolve in time,

the way that they flow over their state space.

And there's, there's a sense in which not only is it much more natural, mathematically, it just works easier.

It looks prettier. It's, it's, it's much nicer.

There's also a sense in which it's, it's meaningless to talk about states of a process, which is constantly in flux.

So limiting ourselves to densities over states as exists in the last, you know, couple of years of the literature, which has in fact, not always been the case.

Some of the early stuff is written in terms of generalized coordinates of motion, which are nothing but a particular way of talking about dynamics and therefore paths.

So there's a sense in which the recent detour into states is very limiting, not only mathematically, but conceptually limiting.

And we can see that directly in the way that it is handling things like non-Markovian processes or processes that change with statistics that are out of equilibrium.

So, so at the, at the very end, we say, you know, we've done all this work.

The, the best next step is to take all of this and, and all of this stuff that we've just done about maximum entropy Bayesian mechanics and put it in dynamical terms.

And this necessitates passing to what we call maximum caliber, which is also due to James originally.

It's, it's, it's a principle of path entropy.

But eventually, you know, the goal is we would be able to write all of this stuff.

And in particular, we'd be able to generalize what we have written into a language that's much more dynamical.

And, and, and Max Cal, we already know, has demonstrated an immense success in treating non-equilibrium processes.

So mathematically, I mean, there's reason to be very optimistic about what we will gain when we generalize the duality that we've talked about previously.

And so it really does seem like it's, it's the ideal thing to do in the very near future.

Awesome.

Let's jump into the sections.

So section one is the introduction.

And just one part that I wanted to highlight that the footnotes are so informative.

There's some of these motifs, like we set out to make it as simple as possible.

And four pages was our limit, but then we ended up innovating this way.

So that's one funny motif.

And then the other funny motif that I see is the super informative footnote often because maybe it's possible to just make the claim stand alone and in direct response to somewhere else.

So that's very exciting and interesting.

Like footnote one covers a lot of the prerequisite fields like dynamical systems theory, calculus, probability, and information theory.

So I found that to be very interesting because it helps understand where we can come from and what kinds of backgrounds we'd want to be learning about.

And then also a very important distinction of the two meanings of the word belief in the probabilistic sense of Bayesian statistics or the propositional or folk understanding as you write from philosophy and cognitive science.

So just thought those were two interesting footnotes and reflected like part of the process and the discourse around the primary text and how that can be clarified in this format.

Feel free to make any overall comments on that or that's just getting to page three.

Well, about the prerequisites, it's a difficult balance to achieve in some sense, right?

Because we don't want to be academic gatekeepers.

No one benefits from gatekeeping.

But these are also highly technical fields.

So, you know, I've been joking for the last few months, if you've never taken a derivative, you're going to have a bad time in this literature.

It's just not the kind of thing that lends itself to, you know, surface engagement in some sense, like it's very and especially with these.

These new developments, I mean, there was Dalton's work that basically shows that the free energy principle is dual to maximum entropy, maximum caliber.

So there's this kind of deep connection to, I mean, effectively the core of contemporary physics.

And there's the there are these new results that show that the free energy principle is asymptotically equivalent to the principle of unitarity from quantum mechanics.

You know, there's there are these new quantum formulations of FEP that have popped up.

So, yeah, I mean, navigating the space comfortably presupposes a lot.

And the good news is that it's hard for everyone.

So there's there's something to learn for everyone.

You know, I'd spent the last three years of my life teaching myself like category theory and gauge theory and stuff, which turned out to be very useful, you know, connecting up with Dalton and doing all of this work.

But now we're launching ourselves into quantum theory, which I'm less familiar with.

So there's something to find difficult for everyone.

But yeah, it I really do think that the fully appreciate going on here.

Some technical prerequisites are important.

Otherwise, we're led to making false assertions like I see this circulate a lot that dynamical systems approaches are contradictory to information theoretic approaches, which is just nonsense from a formal point of view.

Yeah, information metrics naturally reemerge when you're considering, for example, how do you measure distances and state space effectively?

It's just yeah, so the the more mathematical background one comes with, I think the easier it is to integrate into this.

Yeah.

Yes, well, indeed, there's a lot of encouraging and good news and hopefully a community that also values participation and accessibility as well.

So agreed important general points there.

Yeah, before we move on, I'll say something really quickly about this second book.

Just because it's it's a remark that's actually pretty important to me.

In fact, Maxwell is the one who wrote it.

But I didn't even realize, but it is an important clarification that when we talk about beliefs and inference, it's much more general than just an agent or a cognitive object.

The manner in which we talk about inference and we talk about carrying beliefs is this much more general sense of just two random dynamical systems that are coupled and therefore reflect the statistics of each other.

So it could be a sit.

I mean, generically, this is a system estimating the statistics of an embedding environment.

The very important point is any system with any noise and in fact, systems with no noise have statistics.

And so any system that estimates or couples to another system with statistics can be interpreted as mathematical inference.

One system performs inference about the other.

The sort of content that you place on that idea of inference is very case dependent.

So it's not the case that that stones, for instance, are cognitive, but it is very much the case that stones reflect the statistics of their environment simply because these two systems are coupled and their statistics reflect each other.

And so this is what we mean in the sense of talking about agents that carry beliefs simply is about objects that do this kind of statistical estimation, which we would think of as inference.

But it's certainly much more general than simply something like a reinforcement learning agent that actually stores representation of its environment.

It is a representation, but in a much more trivial sense in a general case.

I mean, I would I would add to that by pointing out that, you know, Dalton was saying that we will there were disadvantages to focusing on the density dynamics formulation, which features Markov blankets and the like.

So Markov blankets are great and we like them, but I think focusing on them so much in the literature over the last several years has obscured the fact that the the free energy principle is at core a statement about how, you know, as Dalton was saying earlier, random dynamical systems coupled to each other.

So, you know, the Markov blanket is not some I often hear and read in the literature, the Markov blanket secludes the organism from its environment, but it's not that's not that's not the idea at all.

The Markov blanket is the set of states via which the organism and its environment are coupled.

And in fact, we, you know, we made this case in a paper on a variational approach to niche construction several years ago.

You can basically swap the sensory inactive states of your organism and you get sensory inactive states of an environment.

It is fundamentally a story about how systems couple and synchronize with each other.

It's not a story about how, you know, things that exist are somehow secluded from an environment.

It's really a story about how, yeah, we we we we become a model of our environment, meaning that we are like in tune with engaged with and reflecting the structure of that environment.

This becomes very clear, by the way, when you look at the quantum formulations of the free energy principle, if you take the asymptotic limit of the free energy principle.

So its behavior is time tends towards infinity for classical systems that have space time individuation.

What you get is something like the good regulator theorem.

So I become as much as possible as I can like my environment, you know, module of the space time separation.

I can't be my environment because I'm here and it's there in some sense.

And then what you get in that context is statistical mirroring.

But in in the quantum context, when there is no space like separation between systems, the FEP tends towards entanglement.

So systems that are related in an FEP theoretic way at the quantum level become entangled.

That is precisely what the FEP does conforming to the FEP means that systems become maximally entangled.

So it really is not a statement about, you know, how organisms are separate from their environment.

It's a statement about how organisms are things in general, self organizing systems individuate and couple

simultaneously to their environment and how that's not a contradiction.

Excellent.

So as we rush through section one and head to section two, I just wanted to pull out two really key quotes you wrote in this paper.

We discussed the relationship between dynamics, mechanics and principles.

And then later on page five, you write Bayesian mechanics is specialized for particular systems that have a partition of states and go on to describe it more.

So as we head towards section two, an overview of the idea of mechanics and indeed Bayesian mechanics. Could you unpack a little bit by what you mean dynamics, mechanics and principles?

Sure.

So Mel Andrews, as everyone I think in the literature knows now, wrote this great paper.

The math is not the territory, which challenged us to be very clear about what we meant really when we're talking about the

free energy principle and active inference.

And in there, I think Mel makes a number of contributions, but one of their most important contributions is to differentiate between the FEP as an uninterpreted formal structure and the FEP as it might figure in applications to model specific kinds of systems.

And Mel points out that, you know, the, a lot of the confusion in the philosophical literature has to do with the fact that the, the primary FEP literature often doesn't differentiate between the, just the mathematical structure of the theory.

theory.

And then these models interpreted as representing certain features of, you know, a target system that we want to make sense of.

And that's important because falsification or verification, empirical validation, more generally applies to one kind of thing, but not the other.

Right. So the, the free energy principle insofar as it is a mathematical theory, i.e. like a formal statement is no more falsifiable or subject to empirical verification than calculus would be.

Right. Like you wouldn't, you wouldn't try to falsify the axioms of calculus.

They do or do not apply informatively to certain kinds of systems.

So there's a level at which the core formal results of the FEP are, are, have that status.

They're mathematical results.

Um, and they, we can use this to, uh, you know, do some interesting empirical stuff.

So, uh, what we wanted to do is to differentiate between three kinds of, uh, formal things in the first class of, uh, tools that, uh, no, was outlining.

Um, namely we distinguish between dynamics, um, mechanics and principles.

Um, so the dynamics are descriptive.

They are basically a, uh, a formal description of some systems behavior.

Uh, you might say that like Galilean mechanics is, uh, almost entirely dynamics.

Uh, you know, the, you, you needed someone like Newton to come around and say, well, there are, uh, general,

equations of motion that one can write down to account for, uh, sort of how these, uh, how this description plays out in some sense.

Uh, so we move from dynamics to mechanics, uh, in the same way that we knew from description to explanation from a target phenomenon that is formalized in a certain way.

Uh, that we would like to have an explanation for, to a set of equations of motion that really account for, uh, how those dynamics come to be.

Uh, how those principles, uh, how those principles tell you, uh, how principles tell you why.

Uh, so usually, uh, in physics, we appeal to, um, some symmetry or conservation principle, uh, in order to derive the mechanics or the, uh, a mechanical theory for the systems in question.

So what we're left with is principles, mechanics, and dynamics, all of which are formal theories.

And once you have a mechanics, you can apply it to study specific kinds of systems empirically.

Uh, and that's where empirical verification would come in.

So principles are not falsifiable.

They're just bits of math.

And you use them to write down mechanical theories, which when interpreted in the right context, and when the variables are mapped onto things that are of interest, target systems, then they become subject to empirical evaluation.

Um, so hopefully that, um, you know, uh, clarifies a lot of the model around, uh, around the FEP.

One of the reasons why we spent so long, you know, outlining the idea of mechanics and physics is that this isn't some exotic way of thinking about things.

This is really just physics, just straight vanilla reasoning from physics.

Um, yeah, I'll stop there.

Delphin has probably some interesting things to add.

Uh, no, I think, I think you've covered it.

Um, yeah, it's, it's just that.

There are important formal differences in physics between what we would call a mechanical theory for some object and a dynamical description of some object.

Uh, and so in this tower of things, um, you get ideas, you get theories that sit at different layers in the tower.

This hierarchy ends up being useful in clarifying, uh, this host of different results in the FEP and where one fits into the other.

So it was important to take the time to discuss it.

Uh, but, um, that was that.

And that's the whole point of section two.

Awesome.

So to also move quickly through section two on page nine, you're right.

You're right.

We are concerned here with what we have called Bayesian mechanics.

And then you launch into an example of mechanical theory operationalizing a mathematical principle.

And you provide some classical statistical formulations on page 10.

So could you give us an overview of what the equations on page 10 are showing?

Like what they do for classical physics and what is the analogy or what is the connection that this is preparing us to make with Bayesian mechanics?

Yes.

Uh, well, I, in fact, I have an entire paper, uh, that's about to come out about this, just like building on the analogy.

Um, so I'll, I'll, uh, leave that as an Easter egg for the audience.

Watch out for a classical physics for the Bayesian mechanic paper.

So I think instant classic, it might be some interest.

Yeah.

Might be of some interest, but yeah.

So on page 10, um, we are just giving an example of the, of this kind of formal difference.

So you have, um, some kind of principles in classical physics.

It is the principle that systems, um, are kind of lazy.

You know, they, there is some kind of, uh, an amount of energy that it takes to get from A to B.

Um, and systems don't waste that energy.

Uh, so they take precisely the shortest path from A to B or the path of least energy, um, in, in the more technical sense of the transfer of potential energy into kinetic energy.

So if you have some amount of stored energy, the transfer of that into energy of motion is minimized along the path.

That's a very, um, abstract statement.

And, and there's not a lot that you can actually do with that computationally.

All you can say is, you know, systems, systems, uh, try and get from A to B using the shortest amount of motion possible.

No extra motion.

So, so if you see in figure one, for instance, uh, there are these paths that fly off into sort of these, these very strange fluctuations.

In the classical world, those fluctuations don't exist because classical systems take this, uh, energy minimizing, uh, parabolic path in, in the darker blue.

What is computational about that is that principle of stationary action.

stationary action.

So the principle that you don't waste energy, um, results in a mechanical theory, a law of motion for the mechanics of classical objects.

This is what we would call Newton's law.

Yeah.

Newton's second law in particular.

It's this, you know, everyone knows F equals MA .

It's, uh, force equals mass times acceleration.

So the idea is that, um, if you are such a system, if you minimize the amount of energy,

that you take, then the acceleration of you must obey precisely the force being applied to you.

Uh, and, um, and so this is your, your law of motion.

This is a mechanical theory.

The dynamics that come out of that are able to be discussed when you start putting in actual details, when you plug in, what is the force being applied?

Uh, sorry, that's a dog.

You probably heard it.

Uh, what is, what is the force being applied to the system?

What are the initial conditions for the system?

What is the mass for the system?

All of this extra data starts giving you trajectories.

So an actual model of the system, which is less explanatory and more descriptive.

And this is at the very bottom of the hierarchy where you get this dynamical theory.

That's got all this extra data in it.

Um, so this is, this is very quickly an account of the, uh, that hierarchy.

Awesome.

Let's move into section three and figure two.

So the caption is three faces of Bayesian mechanics.

What are the three faces of Bayesian mechanics?

And what is this figure showing?

Well, um, it's, it's often been suggested that there are like 17,000 versions of the free energy principle.

Uh, and that it's not very clear, you know, what the current version is and all that stuff.

And some of this is fair and some of it isn't.

I think some of it is fair because it is true that often in the technical papers, not sufficient care has been given to, you know, making very explicit which assumptions we're making about the systems that we're interested in modeling.

Um, so what we tried to do in section three is to provide a comprehensive survey of all of the ways that the free energy principle has been articulated.

In the literature.

Um, and so it turns out that there are basically three main ways in which the FEP has been applied, which we're representing here.

So we've already alluded to the first distinction.

Um, the FEP can either be formulated in terms of a probability density over states directly, or, um, as a probability density over paths.

So over trajectories of states that we write in so-called generalized coordinates.

Um, and those are importantly different.

Um, so the, the literature, uh, over the last, I guess, like circa 2009, 2012 to 2019 focused, you know, almost exclusively on the, uh, density over states formulation.

And this is where you get, um, you know, the approximate Bayesian inference lemma that's played, uh, a

pretty critical role in a lot of these investigations.

This is also where you get, uh, like states that play the role of Markov blankets effectively.

Um, whereas the original formulation of the FEP was in terms of a probability density over paths.

So you're not, you're not really so much interested in, in the, the, you know, the, the, the dynamics of, uh, the system just in terms of how probable states are.

What you're actually interested in is, you know, entire trajectories, uh, that unfold over time.

And, um, the, the most general, you know, paths formulation doesn't make a lot of the, um, assumptions that have been criticized over the last little, uh, bit.

So importantly, uh, we don't assume that the, uh, systems being described are stationary.

Um, we don't assume that, uh, you know, and accordingly, we don't assume that the system has a non-equilibrium, a steady state.

Um, and we're now returning to this formulation because it's less restrictive, uh, you know, for reasons that, uh, Dalton could probably detail, uh, better than I can.

Um, so then within the density over states formulation, um, which you get is this kind of mode tracking or mode matching behavior where, um, I mean, essentially what we're saying is that, um, a, a system that conforms to the free energy principle is such that it looks as if, um, the average internal state is tracking the average external state.

Or, or like, uh, matching it.

And the, the relevant difference, um, in that, uh, density over states formulation is that the, uh, external mode that the system is tracking can have dynamics to it or can also not.

Um, and it was important to distinguish, uh, these cases because a lot of the formal analysis.

So for example, our colleagues at Sussex, uh, released this great paper, how particular are the physics of the free energy principle.

And they come to the conclusion that, uh, there's, the FTP doesn't really have anything interesting to say about the class of systems that they examined.

Uh, they are looking at linear systems, uh, that undergo, uh, dissipation effectively.

So, uh, physically these are dampened springs, right?

So, uh, to restate, if you take away the mathematical, uh, baggage, what they're saying is the FTP doesn't have anything particularly interesting to say about, uh, the behavior of dampened springs.

So what does a dampened spring do?

Well, it starts off and then it dissipates back to a fixed point, right?

Like, just due to friction.

Um, so the mode towards which the system is, is flowing has no dynamics to it.

It is a fixed point.

It just dissipates back to a fixed point.

And what they're showing is that there's nothing interesting to say from an FEP theoretic perspective about those systems.

But I mean, the, the counterpoint is sort of the, there isn't interesting, anything interesting FEP theoretically to say about those systems.

There isn't any interesting, uh, external dynamics that, uh, the internal dynamics would be matching.

Um, and a lot of these null results, uh, where we, we, we, we use FEP technology to say stuff about, uh, a system and there's nothing interesting to say.

Look at systems that have fixed modes, fixed external modes.

And so, you know, the, the, the idea is that, you know, the FEP applies vacuously to those systems because there's nothing interesting to say.

Um, so yeah, to, to summarize my, my rant, um, we, uh, we tried to summarize the main applications of the FEP in the literature.

We identified three, um, and strategically, uh, we point out where some commentators may have, uh, been led astray, uh, by focusing on one rather than the other.

So in particular, a lot of this, um, these ideas that the FEP can't handle things like, you know, systems with moving attractors and, uh, you know, systems that don't have like a single mode.

Um, that these are understandable characterizations given that basically since 2000, from 2018 to two, sorry, 2012 to 2019, we mostly focused on this middle class of systems.

So systems that, uh, we're describing in terms of the density over states, uh, that have a dynamic mode.

So it's understandable why people would focus their attention on that.

Um, but the FEP in its most general form doesn't make, you know, perhaps problematic assumptions like stationarity.

Um, and it, yeah, it, it also applies interestingly to systems that have interesting external state dynamics.

Um, so strategically, I think that's what we were going for in this section.

Um, and I, I, I believe this is the only place in the entire literature where this is reviewed so comprehensively in terms of this tripartite distinction.

Um, so, you know, I, I don't mean to be disingenuous and to suggest that, you know, the confusions that have arisen are only due to, um, you know, just, uh, a lack of paying attention.

These things have not been expressed very clearly.

The assumptions have not always been stated very clearly.

And yeah, it, the, um, the form of, uh, the path of least action in each of these things looks very different.

Um, so it, it, we can also understand why it might seem like there are 17 versions of the free energy principle.

Um, but there aren't.

Thank you, Maxwell.

As we speed through this section, I'll just note that formalism one on page 15 is familiar and has to do with the flow of states in the particular partition.

And we've also seen the, um, Helmholtz decomposition shown in figure three in terms of a decomposition on, um, into the, the vertical and the horizontal, the gradient pursuing and the solenoid isocontour aspect of flow.

Those are reviewed in the citations that are provided here.

So let's get into the newer stuff.

Um, in section four, you're discussing some mathematical, mathematical preliminaries on the maximum entropy principle gauge theory and dualization.

But first, yes, Lars.

Moving on to section four, because I thought section three was incredibly illuminating for me personally, and I

think the literature in general,

the literature in general, and having this separation out.

And there was a question that arose for me and working through this that I just wanted to throw out there.

Um, I don't want to distract the journey through the paper too much, but in moving to this, um, you talk about on page 19, um, when the FUP is applied the density over states as opposed to path.

We assume non-equilibrium steady state density.

And then on the next page, page 20, you see, we can view this NESS non-equilibrium steady state density as providing a set of prior preferences, preferences, which lead the particular system, um, to look like it attempts to enact and bring about these preferences through action.

If you then move to the path based formulation where there is no NESS assumption, how do you recover these preferences in, in that formulation?

Or, or, or can you not use that formulation to describe systems that look, are doing actions that look like they have preferences or is there a piece that I'm missing?

Does that make sense?

Um, I can take that one.

So, um, one of the, one of the key, uh, deliverables of this third section is this three faces type thing, right?

So the idea of the free energy principle and, and the mechanical theory that arises from is we would want a mechanical theory, a law of motion for Bayesian inference or for systems that do Bayesian inference.

And what it ends up being is approximate Bayesian inference.

And, and this comes in three flavors mode matching, mode tracking or path track.

This is the, the three faces thing.

So that, I mean, just like you have different flavors of classical mechanics, you have things that move continuously like satellites in orbit, or you have things that move and then stop moving like a bowl thrown through the air, eventually returns to the ground and things that sit on the ground don't move unless they're acted upon.

So there, there are very different flavors of approximate Bayesian inference.

And these three are in particular, the kinds of things that we're thinking about.

So the reason why I give that context is because we can still imagine preferences on the paths of a system.

A system must evolve in such a way that it avoids certain areas of the state space, given that it remains a system.

So, you know, a human can't evolve in such a way that it dips too far away from its allostatic bounds.

There are certain set points and certain bounds around those set points in which I can oscillate, but too far outside of those bounds.

And I'm in trouble.

Likewise, you know, I can have something like a stone that kind of evolves through a state space.

The stone doesn't have preferences per se, but it has a definition.

And so we can read not preferences, but definitions about what a stone looks like in the same sense as we understand what a human looks like as certain preferences that guarantee cohesion.

So things that are definitional of a stone guarantee that same kind of cohesion that just there's a type error if you would identify them.

And what's interesting about that is now you can talk about preferences or you can talk about definitions along the path of evolution of a system.

The stone must stay roughly together. It must stay in this sort of crystalline structure.

If you start hammering away at it, you turn it into a powder.

And this is something that we wouldn't think of as a stone.

So in the same sense, if a human doesn't eat for too long and maybe starves to death, it's a really grim example, but unfortunately, it's one of the better ones.

The better ones you can imagine, you know, you're drifting too far away from this state of being satiated or having sufficient energy to maintain cellular processes and the chemical reactions that sustain life.

And as a result, you know, the preference along your set of paths has been violated.

So I think, yes, there is a sense in which even though we have forgotten a nest density per se, we still have a density over paths, not over states, that this density still expresses something of a definition for the system or a set of preferences for the system without which we don't have a conceptualization of that system.

That's a very long answer to your question, but maybe it satisfies you.

No, that's great. Thank you so much. That was really useful. That makes sense.

Thank you. You can kind of think about it as these constraints force systems to basically pick trajectories that are at the trow of, you know, a preference landscape or a constraint landscape, if you want to think about it.

So you get the same kind of thing as Dalton was saying, but it's not the systems aren't assumed to be stationary.

So they're not, you know, non equilibrium steady state distributions per se.

But there is something analogous.

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The end.

that's pursued in this section is through the lens or at least related to gauge theory so

we're looking at figure four what is gauge theory what is category theory dual and how

is it being deployed in this situation um i guess i'll take that one as well uh figure four you said

yes okay illustration of fiber bundles sections it's a big mega figure oh yeah okay yeah uh well

mega thing i i recognize that name i believe that's how we refer to it internally uh yeah well so

if you ask me um because i am a mathematician primarily uh if you were to ask me or if you

were to ask a mathematician what is the gauge theory uh you would probably get a response roughly like well gauge theory is the study of a particular geometric space called a fiber bundle so there is there is a type of space called a fiber bundle in geometry uh it looks exactly like that figure it's a bunch of fibers pasted onto um something on on the bottom uh what we call a base manifold uh and uh and studying how do things behave in a space like this how do they behave in the sort of fibers and how does that reflect some kind of behavior on the base manifold um these are interesting geometric questions uh that arise in the study of fiber bundles and hence in in what is called gauge theory if you're a physicist i mean it turns out that fiber bundles are the natural uh space for talking about um classical uh gauge theories so the the theory of certain high energy particles um but it's but it's a little bit more general than that uh and and so if you're a physicist this is what you think of you think about um like gauge bosons uh but but this is only because fiber bundles are a very natural way of talking about these things and so it is a little bit more general what we're talking about in the figure four is this idea that uh in this space of fibers uh you can kind of trace out um surfaces and that these surfaces are like probability densities over the base manifold and in fact they aren't just like probability densities they are so you can formulate probabilities over a state space as an object in a fiber bundle and therefore um if you have all the technology available to you you can draw this very nice uh gauge theoretic interpretation of what it means to be uh a probability density and therefore because the object of bayesian mechanics is systems that do inference and so systems that estimate probability densities there is a nice gauge theoretic interpretation to bayesian mechanics that hinges pretty much pretty much on this uh picture um what we would call the the gauge in this situation is the uh choice of this j function so you see you have several different surfaces uh three of them each of which comes with a different choice of j uh so the the idea of choosing a function in the fibers um and and therefore choosing a shape for your surface uh this is what we would think of as a gauge theory and and that's why uh i went down this route in the uh geometry and analysis paper because it ends up drawing a very nice picture of what it might mean to uh engage in constrained inference uh so the the more the less technical version of it is uh so gauge theories and galton correct me whenever i say something that's wildly or even not so wildly inaccurate but gauge theories are basically how one would um how one articulates uh symmetries or conservation laws in contemporary physics physics uh so most physics relies on the idea that um that some transformations that are of interest like the ones uh described by maxwell's equations uh have some kind of symmetry or or that concern i think a slightly better way of going about it would be to say that you know modern physics is broken up roughly into stuff and forces and forces and forces act on stuff and stuff gets acted on by forces um so the interesting thing in gauge theory is you can talk about how a definition of some stuff the shape of some stuff is intimately dependent on the forces acting on the stuff and so this is where we get this idea of we have a a particular shape to our surface we have some you know probabilities over our states but that this is actually defined by the the the choice of gauge this this j function so our assignment of probability to states is actually determined not by probability but it's determined by the constraints on those states certain states are preferable certain states are not unpreferable states get low

[illegible]

That's the first step.

Your second step is then you hit it with this exponential function.

And e^{-x^2} is, of course, a Gaussian density.

So that's all that figure six is depicting.

It's just zooming in on one of the objects in figure four.

Great.

And I thought on page 34 also, the striking result that the splitting of the flow of a system into vertical and horizontal components under the constrained maximum entropy principle is isomorphic to the Helmholtz decomposition of the flow of the autonomous partition of a particular system.

So for those who are caught up with the way that the Helmholtz decomposition was used, for example, in the papers that we discussed in livestream 32 with Markov blankets and stochastic chaos, this is augmenting it or transposing it into another formalization.

And moving on again quickly.

Well, it's worth pointing out that there are some significant advantages to reformulating things in terms of maximum entropy.

I'll let Dalton speak to the more technical parts of this.

But I guess one thing that pops out to me more politically is that maximum entropy and constrained maximum entropy.

I mean, they are just core parts of the physicist's toolbox.

So the fact that we are able to reproduce the Helmholtz decomposition in terms of the way that flow splits up in the gauge theoretic formulation, I think is quite significant.

It gives us an entirely independent line of reasoning arising at the same results.

So, you know, especially due to some work by Martin Beal and colleagues a few years ago, we were not we were no longer sure whether some of these results obtained.

You know, the there had been doubt thrown onto the derivations of the approximate Bayesian inference lemma in some papers.

Life as we know it from 2013, I believe, is the paper that they targeted.

One way of reading this is that independent of all these analyses and independent of whether Beal and colleagues are correct about their criticism of the 2013 paper.

This is an independent derivation that just follows from uncontroversial mathematics.

And we're able to re-derive basically the main core portions of the formal results that Carl had derived.

So the approximate Bayesian inference lemma, the Helmholtz decomposition.

And I mean, in some sense, the magic is that we understand now why these arise to begin with.

There's there's yeah, I think the the the maximum entropy interpretation in some sense, especially in the gauge theoretic formulation, in some sense tells us, well, why?

Why does approximate Bayesian inference work just full stop?

And these sets of results explain, you know, effectively why this has to be the case.

Only my facilitator hat prevents me from being that excited.

Section 5.4 also explored more in the previous citation.

So the four things that I wanted to just touch on as briefly as people wanted as we close are the three philosophical areas that are introduced in Section 6.

And then, of course, just the appetizer of G-theory.

So in page 38, Section 6, the philosophy of Bayesian mechanics, 6.1 addresses clarifying the epistemic status of the FEP.

So what were you aiming to bring people's attention to and address in 6.1 with this philosophical question?

Well, I think this basically summarizes the point about dynamics, mechanics and principles that we were making earlier.

You know, so the FEP has been described variously in various and sometimes very different ways, even by like its core proponents.

So Carl introduced the FEP to the world as a theory of cortical function circa 2005 and then as a unified brain theory in 2010, which led a lot of people.

And I think not to this is in bad faith.

It led a lot of people to say, well, OK, so this is a theory and so it should be making predictions and therefore it should be empirically verifiable.

So I think, you know, the way that we talked about it led to a lot of confusion and it's been described as, you know, a framework, whatever that means by myself included.

Right.

So mea culpa.

We called it the active inference framework for a long time without really being specific as to what that meant.

We kind of met all this this cluster of results that kind of agglomerate around the FEP.

Raja and colleagues have called this a trick, kind of like a variational autoencoder trick in machine learning.

So it's a it's a trick.

If you can write a Markov blanket for a system, then you can make you can write down a model that looks as that says that the system looks as if it's performing Bayesian inference.

It's also been called a physics, a physics of sentient systems, a formal ontology.

So the point is to say, well, everyone is kind of right.

You know, these are all in some sense valid assessments.

And Mel is really the one who clarified things initially, I think, by distinguishing formal structures from interpreted empirically evaluable models.

And yeah, I think the first part here is to say, well, we can make sense of this model by appealing to this tripartite distinction that we've outlined.

And so by distinguishing dynamics, which are descriptions, formal descriptions of some behavior that we want to get to mechanical theories that explain how the principles that explain why.

And then what we're calling, I think, empirical applications, which are basically these formal models plus an assignment or an interpretation or evaluation of these models in terms of the features of target systems that we want to explain.

Yeah, so when they're applied in that way, mechanical theories become empirical theories, as we write, in the ordinary sense.

But yeah, so hopefully this kind of clarifies everything around falsification, how this thing is used, and

importantly, why this is just standard physics.

I mean, this isn't all that spooky in terms of the use of mathematics to make sense of physically relevant aspects of our universe.

So that's what we were trying to do in that section.

Great.

Really clear.

6.1, it reads to me like, it's the tip of an iceberg with a nice landing and takeoff strip.

And yes, it's a big iceberg, but it is an iceberg, and there is a tip that makes sense.

6.2 on page 39, Ilan Vital and the FEP.

What does this French linguistics have to do with the FEP, Maxwell?

Well, Dalton might have some pretty interesting remarks to make.

But just briefly, I mean, I find it kind of ironic that, you know, the people who are most critical of FEP right now, like folks from the inactive tradition and folks from the ecological tradition, sometimes frame their own research as trying to, you know, struggle against dualism in some sense.

But our belief is that they are intrinsically dualistic frameworks.

So Ilan Vital is a French term with vitalism.

I don't know how familiar you are with like the history of biology, but before the 20th century of the discovery of DNA, most serious physicists believed in something like Aristotle's entelechy.

So like you had to have some kind of physical mechanism for purposiveness.

And so Ilan Vital was this kind of vital force that they believed, the vitalists believed, animated living systems and differentiated them from non-living systems like stones and rocks.

And we believe that one of the consequences of the perspective that we're adopting here and building on, you know, especially the work of Mike Levin, who's done amazing, made amazing contributions in this field.

Is that it self-organization and cognition and life and these things that we like to point out as being distinct from inert matter are actually are more on a continuum.

You know, the Bayesian mechanics says that basically anything that's a thing will exhibit some features of what we think of as cognition.

So it is a massive, massively deflationary and absolutely resolutely monistic perspective that we're advancing here.

Yeah, which says, you know, the trying to look for the boundary between living and non-living systems is bound to fail in some sense.

There is a continuum.

These processes that we're interested in are also evinced by rocks and by tornadoes and by crystals.

And it is a question of degree to which systems act as cognitive systems.

Yeah.

I'm sure Dalton has more interesting things to say about that.

Yes.

On 6.2, what else would you add there, Dalton?

Well, maybe not more interesting, perhaps more interesting things to say.

But, well, yeah, I mean, I think this section is mostly just a recapitulation of footnote 2, in a sense.

Meaning, you know, we made this statement that inference, per se, representation of some statistics of something else,

is this very broad, very general thing.

And any system that is not a closed system does inference.

So what that means is, you know, under the free energy principle, and at least what we've written in that paper,

you know, what we've worked out so far, there isn't a way to talk about cognition versus non-cognition.

It's more of a spectrum of cognition.

There are certain systems that are more cognitive.

This is something that Carl might call temporal depth.

So there are systems that are very good at planning and executing actions.

And there are systems that are not, you know.

So there's a very clear difference between humans and stones, for instance.

But that difference isn't in one does inference and another doesn't.

It's that one does a bit more complicated inference.

It has better or more accurate representations or more representations, full stop maybe.

And it's better at using those representations to do action.

But both stones and humans are estimators of the statistics of the processes around them, simply by mathematical construction.

And so there's this very uncontroversial statement, or at least it shouldn't be a controversy, that systems, things, in the very most general sense, exhibit this kind of representation-based inference that we would typically identify as cognition.

So that was the motivation for talking about that.

And then Maxwell tied a very nice bow on it with this great history of a debate around very similar questions.

But yeah, it was this idea of a spectrum of cognition.

Awesome.

And citation 92 there is the 2018 paper answering Schrodinger's question.

So many years ago, the other Maxwell et al., who raised that question and sparked really a phase transition in the discourse.

And that was also like one of the papers that really drew me in and got me excited.

And so it's amazing to see what other questions and answers have been risen.

6.3 and then G.

So 6.3 is a slightly longer section, and it's on maps and territories.

So just what is there?

I'm really glad that we were able to write this.

So I never found this particularly confusing, but it has led to a lot of confusion.

So in particular, there's this entire paper by Thomas Van Ness that just is just confused about this issue.

I forget what it's called.

It's a living life model.

Anyway, so folks have said, well, so is the free energy principle just a description that we are?

Is it a model that we are using to describe interesting features of the world?

Or is it saying that self-organizing systems actually are models of the world?

And so, you know, I never particularly found this problematic because I, so this phrase in the paper, I think, sums it up correctly.

The free energy principle is in some sense a map of that part of the territory that looks as if it is a map.

And so Thomas Metzinger actually retweeted our paper and pointed to, it's called this nested representationalism, which I found interesting.

So, yeah, there are these kind of two distinct levels at play, but there's no foul play here because the FEP model is very explicit about which parts of the formalism pertain to our map of the territory and which parts of the formalism pertain to the map that self-organizing systems are constructing according to our map.

Yeah, so I think it's important to point out that, you know, there isn't a contradiction here.

There are just these two probability densities that are at play in the construction, the Q density and the P density.

And you can understand the generative model or the P density as basically our model of the self-organizing system in terms of, yeah, joint probability density over its state space or potentially its space of paths.

So in that sense, you know, the self-organizing system being its own best model really means we have a model.

We can explain the system's dynamics and the system's potential is formalized, you know, as a generative model, as a joint probability density.

It just so happens that part of the variables that are described by that joint probability density play the role of parametrizing additional probability densities over another subset of variables described by the generative model.

So there are really these two, I think, unambiguously different senses of being a model that are at play.

And yeah, I don't think this is a sleight of hand.

And hopefully we have clarified.

Yeah, what what exactly is at stake here?

I think there are some nice connections to the maximum entropy formulation.

So as Dalton articulated in the paper, at least, you said as much in discussion today.

I think you by by flipping to the perspective of maximization of entropy, we we effectively recover our own position as modelers trying to model the system.

So the the the the the prison through maximum entropy is adds this additional clarification that, yeah, we can entirely recover the perspective of a modeler using maximum entropy modeler to model a free energy type system by invoking this duality of perspectives.

And I think that's really important.

Then there's this other piece here, which says that the the maps that we construct in FEP and in Max and both adhere to a kind of epistemic minimalism.

So maximum entropy inference is basically saying I'm going to give in some data set.

The probability density that most that is the one that best explains the data set is the one with the highest entropy.

What this means is that we are building in to that probability distribution as little prior information as possible. It's like starting from the point of view of maximal ignorance, given a few constraints, what is the underlying probability density?

The FEP has a similar starting point for maximal ignorance, which says I can't go beyond the blanket like I am stuck with the data that I'm generating and with the way that I probe the world, the data generating process to generate my data.

And there is no going beyond that, you can't go beyond the blanket unless you break the blanket, but then you're just introducing a new blanket.

So, for example, you know, I can't see in your brains.

I could put you in an fMRI machine and break the blanket, but then I'm really introducing another another blanket, you know, which is this new set of data that I have to continue making inferences from.

So, I mean, then the FEP and Maxent kind of emerge as these epistemically minimalistic or, you know, they kind of emerge as the way that you would construct a map of the part of the territory that acts as a map from a perspective from a starting point of maximal ignorance.

Great. And I think that's connected in the active inference textbook to figure 9.1, the meta Bayesian perspective.

And there it's not described in the context of maximum entropy, but it's something very similar with this nested representationalism on any other philosophical notes before we go to section seven.

Yeah, that's exactly right. That's exactly right. Yeah. Yeah. Yeah. I love this book.

Yeah.

And that's anything to add before we launch into G theory.

No, I think you raised an interesting point.

So we can talk about, you know, when we use the FEP to model a system, what we're doing is forming a model of the system where the system forms a model of its environment.

And so there is this nested representationalism. What's interesting about the maximum entropy point of view is you can kind of dualize it.

You can you can take this sort of mirror image of this thing and you can ask, OK, well, we're not interested in a model of the system modeling something.

All we want is a model of the system. So what does the system look like when it's creating that model?

Not what does the system's model look like?

So this is this is the dualization aspect of things. And so you can talk about what and when you dualize, you can talk about things like constraints on agents states rather than the predictions and actions of the agent.

So it's a bit of a different viewpoint. It's just emphasizing different levels of the nest, so to speak, which is which is kind of interesting because you do get both of these perspectives for free as soon as you commit to this relational symmetry of an observer in an environment and an agent in an environment and how those two things do model each other.

And that's what you get when you dualize things. But that's yeah, that's that's it for me on that section.

I think Maxwell pretty much covered a lot of the important points.

I mean, I will say that, you know, this this probably is not obvious until you really look at the mathematical formulation and notice that there are two, you know, as I as I wrote in that paper from a few years ago, the FEP is a tale of two densities.

And you really need to like take take a look at the math to understand that there isn't a this isn't like a case of reification.

This is just formulated. So it isn't so simple that you could just put it in a direct and a simple graph and say, hey, look, here are these two.

But definitely if you take the time to work through the formalism, there are clearly these two different senses of being a model that are at play, that are cleanly distinguished and I think are actually not all that problematic.

Awesome. It's almost like thinking through other minds highlighted some of the social consequence and theory of mind and behavioral aspects of certain types of cognitive entities.

But systems thinking of thinking of thinking of thinking of thinking through other infers, that type of dual view from the inside view from the outside is again what you're suggesting is enabled by that kind of a commitment to this foundations.

And of course, it could be a whole other guest stream and perhaps should be on G theory.

But as we close here, what can we say today on 613 2022 about G theory and what is exciting you all about how you're going to continue forward?

Do you want to take this one, Dalton?

Yeah, sure. So, I mean, like I mentioned earlier on, there's already a lot of really promising stuff in the literature about path integrals and path entropy.

And the kind of power it has to deal with systems that we would normally kind of balk at.

And so it's already very promising. There's reason to be optimistic about how this extension will benefit us in the future.

So G theory itself is, Maxwell coined this term to describe the extension of this duality from mode matching and max cent, max cal, which will involve precisely this.

We're trying to formulate the free energy principle in its full generality as dual to entropy in its full generality on paths and so forth.

And so the reason why this is so exciting is because already there's a very clear path towards talking about systems that are non stationary.

So systems whose statistics change, more complicated forms of inference, more complicated forms of constraints and preferences.

And so it seems like already it's a much more faithful description of the kinds of systems that we want to talk about and extends to things like more complex systems, things like humans potentially.

So this is really the end goal is to do this dualization to now build an FEP dual model of something like a brain, which we know models its environment in this very complicated way.

And so the kinds of representations it does are very complicated.

So how do we model that?

That should arise out of maximum caliber and therefore out of G theory.

And in principle, you know, eventually we'll have a very faithful, very foundational picture of complex systems.

But this is a very long term goal in the short term.

We just want to get the maps to work.

To connect that a bit to Lars's question earlier.

What does it mean to have a non stationary density over paths?

Well, I mean, what we're interested in is the way that the density changes as a function of the dynamics so that you really have like these moving surfaces over a space of paths or a space of states.

And to really capture all of that kind of time dependency requires moving from Maxent to MaxCal from states to paths and all of these kind of more fine grained.

You know, wandering blankets and moving attractors and all of that good stuff should fall fairly naturally out of G theoretic formulation of this stuff.

I like how within FEP, as we know it, we have variational free energy F and then expected free energy takes us into the future and planning and that's G.

And then we see a little bit of like free energy principle F and then G is the next letter.

So that's a funny little would know dual analogy.

Well, if anybody has any closing thoughts, it would be a great time for that.

But just wanted to say that there's a lot of appreciation in the chat and a lot of excitement around this work.

So we hope that you just continue acting and inferring and serving.

That's kind of you to say.

Yeah, thank you very much for the invitation to discuss our work.

This has been great.

Yeah, we we look forward to, you know, exploring the future with this whole community and thanks for your time or attention, your energy.

Very, very positive.

Always a pleasure to be on Actimflab.

And yeah, thanks for the guest stream.

Yes, I can only echo that.

It's been a pleasure.

So thank you.

Thank you.

Thank you, Maxwell and Dalton at all.

Thank you, Lars, for joining.

And until the next time.

Bye.

May the gradient descend ever in your favor, my friend.

And with yours slash ours.

Bye.

Take care.